

Beyond “1 GB” Graphics

If you look at the specs of a laptop, the two prominent things mentioned are the GPU model number and the amount of memory dedicated to the GPU and most dealers would care about just that much disregarding the fact that there is so much more to it. First things first – unlike processors, both AMD and NVIDIA use higher numbers in each series to denote better GPUs. So an NVIDIA GeForce GT 635M is going to be better than NVIDIA GT 630M and you can almost depend on the processor numbering in each series to rank one GPU better than the other. But then, do we settle down with model numbers alone? Nay, there be more to it.

Before we talk about which ones are good, it would be wise to ponder over the question “how does a 900 MHz GPU outperform a 3 GHz CPU in graphics performance?” Let us consider one of the best mobile processors available in the market – Core i7-4930MX Extreme Edition and one of the best mobile GPUs in the market – NVIDIA GeForce GTX 780M. The former runs at 3GHz clock speed normally and can go up to 3.9 GHz with turbo-boost while the latter



Memory size is not the only spec that matters

runs at 823 MHz clock speed. If you look at just these two numbers, the graphics card looks like a kid. Expose the fact that the processor has only 4 cores (with the capacity to churn 8 simultaneous threads) while the graphics card has 1536 such processors running, you get the large part of the picture.

Let us do a very rough calculation of speeds with assumptions favouring the CPU:

- i. Suppose the CPU is able to work on 8 threads at a time at 3.9 GHz (that wouldn't happen because there are actually only 4 cores to do the work). The total number of computing cycles it is running per second = $8 \times 3.9 \times 10^9 = 31,200,000,000$.
- ii. The GPU is working at 823 MHz with load on all its 1536 compute cores. The total number of computing cycles it is running per second = $1536 \times 823 \times 10^6 = 1,264,128,000,000$

Even with such insane favours to CPU, the GPU is about 40 times faster